Geometrically Inductive Self-Supervised Learning for EEG: A Design Study of Spatial Attention Across Sessions, Subjects, and Montages

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Abstract

I propose, implement, and evaluate a self-supervised EEG encoder whose spatial attention is conditioned on a geometric descriptor g_{ij} — built from the 3-D scalp positions of each electrode pair — instead of a learned per-electrode codex. The motivation is that EEG non-stationarity, across sessions, subjects, and montages, is largely a story of electrode geometry shifting, and a codex of per-electrode embeddings cannot absorb that shift smoothly because its parameters are indexed by electrode identity rather than by structure. Crucially, a codex model *cannot even be evaluated* on a montage absent from pretraining: it has no embedding slot for unseen electrodes. The training recipe (momentum-encoder alignment with masked-patch reconstruction) follows EEGPT [14]. This report covers the **v2** experimental regime: a leave-one-montage-out, mixed-corpus pretraining design that directly tests zero-shot cross-montage transfer — the property the proposal flagged as the central promise of the geometric approach. The only baseline that can also transfer zero-shot is a channel-independent encoder (`chind`), which discards spatial structure; the codex baseline is deferred precisely because it cannot run on an unseen montage. The headline finding is **positive**: pretrained on a balanced two-montage corpus of 156 subjects (PhysioNet MI + Sleep-EDFx) and evaluated zero-shot on the held-out 3-channel BCIC-2B montage, the geometric encoder reaches 0.545 BAC and beats the channel-independent baseline (0.509) by a statistically significant +0.036 (paired t-test $p = 0.023$, Wilcoxon $p = 0.027$, 7/9 subjects); the reduced position-only descriptor also beats the baseline significantly (+0.022, $p = 0.013$, 8/9 subjects). At a smaller balanced 9-subject-per-dataset scale, the geometric encoder leads `chind` on all three held-out montages but the gaps are within noise. I frame this as an informative design study: geometric conditioning delivers a real, significant zero-shot transfer advantage over the only comparable baseline once the pretraining corpus is large enough, on the one held-out montage that the three available datasets allow to be tested at scale.

Code: github.com/tianxin-scu/geometric-eeg-ssl.

1 Introduction

Non-invasive BCI and EEG. Brain–computer interfaces (BCIs) offer transformative potential for individuals with motor impairments. Invasive modalities such as ECoG and Utah arrays deliver the highest signal fidelity but require neurosurgery and are inaccessible to most candidates. Electroencephalography (EEG) is the natural non-invasive alternative — safe, affordable, widely deployed, and capable of millisecond-scale recording — but its scalp signals are noisy, mixing many cortical sources through volume conduction and contaminated by muscle and ocular artifacts. Better representation learning on EEG has a direct ethical payoff: improved models extract more usable signal from the non-invasive modality, lowering the threshold at which a patient needs to accept surgical risk to achieve functional BCI performance.

EEG is non-stationary, by default. Any deployed EEG model faces a signal that shifts continuously along three axes, none of which is an experimental edge case: (i) **cross-session drift** — recordings of the same subject on different days differ because the cap sits differently each application (several-millimeter placement shifts are typical) and because the subject's physiological state varies (alertness, fatigue, electrode impedance); (ii) **cross-subject anatomical variability** — head size, cortical folding, and skull thickness differ across individuals, so `C3` on one subject does not record from the same generators as `C3` on another; and (iii) **cross-montage layout differences** — clinical and consumer devices deploy 2 to 256 electrodes in heterogeneous layouts, and a model trained on a 64-channel research montage has no principled way to process a 3-channel headset without retraining. The placement, anatomical, and montage components of these shifts share a common structure: the geometry of the electrode array, relative to the underlying cortical sources, has changed. A model whose spatial representation is conditioned on geometry should absorb that change smoothly; a model whose spatial structure is encoded in per-electrode parameters cannot. Physiological state is a separate non-geometric axis of session variability and is not addressed by the architectural change proposed here.

The shared blind spot of EEG foundation models. Recent EEG foundation models — BENDR [10], BIOT [15], LaBraM [8], EEGPT [14] — bring self-supervised pretraining and linear-probe evaluation to EEG with strong downstream results, but they fall into two camps that both fail to handle EEG's natural non-stationarity. The first camp uses a learned *codex*: a table of per-electrode embeddings $\{c_i\}$ indexed by electrode identity, added to patch projections before any spatial mixing (EEGPT, LaBraM). The codex embedding for `C3` is a single vector fit to the average `C3` placement across training subjects; it cannot express that today's cap has shifted 5 mm or that this subject's head is larger than average, and *an electrode absent from the pretraining set has no codex entry at all* — so the model simply cannot be run on a new montage without inventing one. This is the classical *transductive* setting [5]: parameters indexed by node identity, no generalization to unseen nodes without retraining. The second camp removes per-electrode structure entirely: BIOT [15] tokenizes each channel independently, robust to variable channel counts but discarding the 3-D scalp coordinates that every EEG system provides as side information. Neither camp uses geometry as an input.

Contributions. I implement and evaluate the architecture committed to in the project proposal, and in this v2 regime I focus the evaluation squarely on the cross-montage transfer the proposal called the headline test. Concretely: (i) **an architecture** in which per-electrode codex embeddings are replaced by a shared function over g_{ij} inside the attention computation, with no per-electrode parameters and an optional masked-reconstruction head — making the encoder *inductive* over electrode sets, and therefore runnable on any montage with known coordinates; (ii) **a leave-one-montage-out, mixed-corpus pretraining protocol** that trains on two datasets and evaluates zero-shot on the held-out third, isolating cross-montage generalization as the dependent variable; (iii) **a fair single comparison** against the only baseline that can also transfer zero-shot — a channel-independent encoder (`chind`), trained under identical conditions. The transductive codex is deferred — not because it is weak, but because it *cannot be evaluated zero-shot at all*, which is itself the argument for the geometric design; (iv) **a descriptor ablation**: I fix the injection to G3 (geometry at both attention score and value, the strongest configuration in my earlier within-family ablation) and ablate the descriptor content — the full g_{ij} vs. a position-only variant; and (v) **an honest, positive headline.** At sufficient pretraining scale the geometric encoder beats `chind` on zero-shot transfer by a statistically significant margin; at small scale the advantage is directionally consistent across all three held-out montages but not individually significant. I report both plainly.

2 Problem Statement

Formal setup. Let $\mathbf{X} \in \mathbb{R}^{M \times T}$ denote a multichannel EEG recording with M electrodes and T time samples, and let $\mathbf{p}_i \in \mathbb{R}^3$ denote the known 3-D scalp position of electrode i, taken from the recording montage's digitized coordinates or a standard template when digitization is unavailable. Self-supervised pretraining learns an encoder $f_{\boldsymbol{\theta}} : \mathbb{R}^{M \times T} \to \mathbb{R}^d$ from unlabeled recordings; pretraining succeeds if a linear classifier trained on top of $f_{\boldsymbol{\theta}}$ transfers across diverse downstream tasks. This linear-probing protocol is the standard evaluation framework in EEG foundation-model work [8, 10, 14, 15], and I adopt it here.

What this project isolates. The open design choice I focus on is how the spatial encoder represents inter-electrode relationships. Existing foundation models are split between codex-based spatial encoders (transductive in electrode identity) and channel-independent encoders (no spatial structure at all). I propose a third option: a spatial attention function conditioned on a geometric descriptor g_{ij} (Eq. 1) built from electrode coordinates, with no per-electrode parameters, so that the same shared function applies to any electrode pair given only their 3-D positions — making the encoder inductive over electrode sets in the sense of [5]. The practical consequence this report tests is direct: such an encoder can be applied, frozen, to a montage it never saw in pretraining.

Research question.

> *Does conditioning the spatial encoder's attention on electrode geometry — rather than on a learned codex of electrode identities — produce representations that transfer zero-shot to an unseen electrode montage, beating the only baseline that can also transfer (a channel-independent encoder)?*

This is a *design study*, not a SOTA competition. I do not ask whether my model outperforms LaBraM or EEGPT trained at scale; I ask whether, controlling for parameter count and pretraining data, geometric conditioning provides a cross-montage transfer advantage over the comparable baseline — and whether that advantage is large enough to be statistically real.

3 Related Work

3.1 EEG Foundation Models

Recent EEG foundation models share a common pipeline: large-scale self-supervised pretraining followed by linear probing or light fine-tuning. BENDR [10], BIOT [15], LaBraM [8], and EEGPT [14] collectively show that this recipe transfers across motor-imagery and clinical EEG tasks. A shared limitation remains: codex-based models encode space through per-electrode identities (transductive), while channel-independent models omit spatial structure entirely. Montage heterogeneity — 2 to 256 channels in varying layouts — is the deployment bottleneck this work addresses.

3.2 Self-Supervised Learning with Momentum Encoders

Momentum-encoder learning from MoCo [7] and BYOL [4] provides stable targets without requiring hard negative mining, which is especially attractive in EEG. MAE [6] adds a reconstruction objective that encourages representations to preserve recoverable signal structure. I adopt this validated combination (alignment + optional reconstruction), as also demonstrated in EEGPT [14], and focus my novelty on the spatial representation rather than inventing a new objective.

3.3 Inductive Representations and Attention over Relational Structure

The framing of this work — replacing identity-indexed parameters with a shared function over structural inputs — inherits its vocabulary from the inductive node-embedding literature. GraphSAGE [5] drew the explicit distinction between *transductive* embeddings (fit per-node, fixed graph) and *inductive* aggregation functions (shared parameters over node and neighbourhood features), generalizing to nodes unseen during training; that distinction is exactly the one I draw between codex-based and geometry-conditioned spatial encoders. GAT [13] showed that attention is a natural way to implement such an aggregator, with attention scores conditioned on relational features rather than fixed by graph topology — the design pattern I adopt inside the spatial Transformer. EEG connectivity studies independently document that task-relevant information is partly relational rather than purely per-channel [2]. I emphasize that this is a Transformer with a geometry-conditioned attention bias; it does not construct a sparse graph or perform message passing in the GNN sense, and full $M \times M$ attention is well within budget at the electrode counts considered ($M \leq 64$).

4 Proposed Solution

4.1 Architecture Overview

Figure 1 summarizes the model. The single architectural choice that distinguishes this design from existing EEG foundation models lives inside the spatial encoder: how electrode-specific information enters the attention computation, illustrated in Figure 2. All other components — patching, momentum updates, masking, temporal contextualization, and loss — follow standard practice [4,6,14]. The v2 backbone recomputes geometry tables per dataset per step, because the montage — and therefore g_{ij} — changes between corpora within a single mixed-corpus pretraining run.

4.2 The Geometric Descriptor

For each ordered pair of electrodes (i, j) with 3-D scalp coordinates $p_i, p_j \in \mathbb{R}^3$, the v2 headline descriptor is

$$g_{ij} \;=\; \big[\, p_i,\; p_j,\; p_i - p_j,\; \|p_i - p_j\|_2 \,\big] \;\in\; \mathbb{R}^{10}. \tag{1}$$

It combines **absolute positions** (p_i, p_j), **signed displacement** ($p_i - p_j$, distinguishing anterior/posterior and left/right), and **distance** (a scalar proximity term). This extends the minimal 4-D descriptor of my earlier (v1) work — which carried displacement and distance only — by adding the two absolute positions, so the shared MLP can localize a pair on the scalp, not merely relate the two electrodes. The descriptor is asymmetric in (i, j), so attention can express directional preferences. Because g_{ij} is a function of coordinates alone and contains no electrode-identity parameters, the same descriptor space is defined for any montage with known scalp coordinates — including montages absent from pretraining.

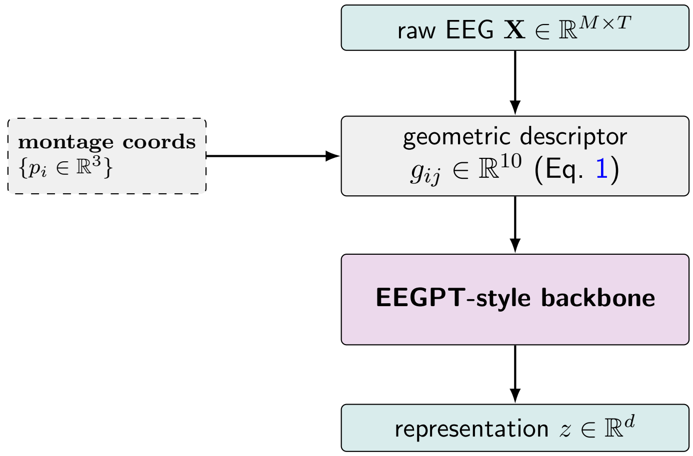

Figure 1: **Architecture overview.** My contribution is the middle block: raw EEG plus the montage's electrode coordinates yield a geometric descriptor g_{ij} that conditions the spatial attention of an otherwise standard EEGPT-style backbone. The codex of per-electrode embeddings used by EEGPT is removed; g_{ij} takes its place as the spatial inductive bias. Coordinates are side information read from the recording montage — digitized when available, otherwise taken from a 10–20 template.

Coordinates are computed in a **fixed scale** from each dataset's channel names, with *no per-montage normalization*. This is critical for cross-montage consistency: the encoder must see the held-out montage's coordinates on the same scale it saw during pretraining, or the transfer is confounded.

Descriptor ablation (pos_only). I also evaluate a reduced descriptor $g_{ij} = [\,p_i, p_j\,] \in \mathbb{R}^6$ (absolute positions only, dropping the displacement and distance terms). This asks whether absolute position alone carries the cross-montage signal, or whether the relational displacement/distance terms are doing the work.

4.3 Geometric vs. Channel-Independent Spatial Attention

Figure 2 contrasts the geometric encoder with the identity-indexed codex. Writing the standard dot-product attention as $\alpha_{ij} \propto \exp\big((W_Q x_i)^\top (W_K x_j)/\sqrt{d}\big)$ with output $y_i = \sum_j \alpha_{ij} W_V x_j$, geometry can enter at the score, the value, or both:

G1 (score only). An additive per-head geometric bias,

$$e_{ij} = (W_Q x_i)^\top (W_K x_j)/\sqrt{d_h} + b_h(g_{ij}), \tag{2}$$

where b_h is a shared MLP $\mathbb{R}^{10} \to \mathbb{R}^H$ (one output per head). Geometry decides *which* electrodes attend.

G2 (value only). Standard dot-product score; geometry-augmented value:

$$v_{j\to i} = W_V x_j + W_{V,g}\, g_{ij}, \qquad y_i = \sum_j \alpha_{ij}\, v_{j\to i}. \tag{3}$$

Geometry colours *what* messages carry, not who talks.

G3 (both). G1 score-side bias and G2 value-side addition, with separate projections of the same g_{ij}. In an earlier within-family ablation (v1) G3 was consistently the strongest of the three injection points, so in v2 I **fix the injection to G3** and instead ablate the descriptor content (Section 4.2).

Channel-independent baseline (`chind`). The same backbone with no geometric term and no per-electrode parameters: standard dot-product attention only. Because `chind` carries no electrode-identity parameters either, it too can be applied to an unseen montage — which is exactly why it is the right baseline for a zero-shot cross-montage test.

4.4 Self-Supervised Objective

The encoder is supervised by the combination of momentum-encoder alignment and masked-patch reconstruction:

$$\mathcal{L} = \mathcal{L}_A + \lambda_R \mathcal{L}_R. \tag{4}$$

$\mathcal{L}_A$ is the alignment term, $-2 \cdot \cos(\text{pred}(z_{\text{online}}), \text{stop\_grad}(z_{\text{momentum}}))$ averaged over batch and time. $\mathcal{L}_R$ is masked-patch reconstruction MSE, applied only at masked time steps; the reconstruction target is per-channel. I default to $\lambda_R = 1$ with 50% temporal masking (v2 uses temporal-only masking; spatial masking is dropped). The momentum encoder is a non-trainable EMA copy of the online encoder, processing the unmasked input. The EMA factor follows a cosine ramp $\tau(k) = 1 - (1 - \tau_0)\frac{\cos(\pi k/K)+1}{2}$ with $\tau_0 = 0.996$ and $\tau_K = 1.0$, matching the EEGPT recipe.

4.5 Baseline and the Deferred Codex

The comparison baseline in v2 is the channel-independent encoder of Section 4.3, trained under identical conditions. The **transductive codex baseline is deferred** in this report, for a principled reason: a per-electrode codex has no embedding for the held-out montage's electrodes, so it cannot be evaluated zero-shot without a hand-designed fallback (random initialization, or copying the spatially-nearest pretraining electrode). Those fallbacks themselves smuggle geometry back in at transfer time, which muddies the very comparison this regime is built to make. The codex variant, and its fallbacks, are scoped as future work (Section 11).

4.6 Why This Design Should Help (the Predicted Advantage)

EEG non-stationarity — across sessions, subjects, and montages — is largely a story of electrode geometry shifting. A model whose spatial attention is a function of geometry absorbs that shift smoothly; a model whose spatial structure lives in identity-indexed parameters cannot even be evaluated on a new montage. Whether the predicted advantage appears empirically — and whether it survives a paired significance test — is what the v2 experiments answer. Unlike the earlier pilot regime, the result here is *positive on the powered test*; see Section 7.

5 Experiment Design

The v2 experiment is a single, focused design: **leave-one-montage-out, mixed-corpus pretraining, with zero-shot linear-probe evaluation on the held-out montage.** This isolates cross-montage transfer — the proposal's headline promise — as the dependent variable, and drops the in-distribution and cross-session ladder rungs of the earlier regime (which had shown no separation and are archived with the v1 results).

Table 1: Leave-one-montage-out splits. Three pretraining runs per variant.

Tag	Pretrain on	Held out (zero-shot eval)
no_sleep	PhysioNet MI + BCIC-2B	Sleep-EDFx
no_bcic	PhysioNet MI + Sleep-EDFx	BCIC-2B
no_phys	BCIC-2B + Sleep-EDFx	PhysioNet MI

Variants. `geometric` (G3, full 10-D g_{ij}), `geometric_pos_only` (G3, 6-D position-only descriptor), and `chind` (no geometry). Every variant is trained under identical conditions — same architecture, optimizer, schedule, masking, loss, and EMA schedule — so only the spatial encoder differs.

Two scales. (1) **Balanced n9.** Subjects capped at 9 per dataset (BCIC-2B has only 9 total — that hard ceiling sets the shared budget, so every dataset contributes equal subject-level diversity). All three splits × three variants give the directional table in Section 7.1. (2) **Scaled n156.** The `no_bcic` split scaled to 78 subjects each from PhysioNet MI and Sleep-EDFx (a balanced 156-subject corpus), evaluated zero-shot on BCIC-2B. This is the *only* split that can be both balanced and large: BCIC-2B caps any corpus that *contains* it at 9, so the one large balanced corpus is precisely the one that holds BCIC out. This run is the powered headline test (Section 7.2).

The pipeline is three scripts: `scripts/v2_pretrain.py` (mixed-corpus pretraining per (variant, split)), `scripts/v2_probe.py` (frozen-backbone zero-shot linear probe on the held-out montage), and `scripts/v2_results.py` (aggregates the per-checkpoint JSONs into the headline tables under `results/v2/` and `results/v2_n78/`).

6 Experimental Setup and Datasets

6.1 Datasets

I use three open EEG datasets accessible through MNE-Python [3] and MOABB [1].

Table 2: Open EEG datasets used in this project.

Dataset	Paradigm	Subj.	Ch.	Rate	Role in v2
PhysioNet MI [11]	Motor imagery (4-class)	105	64	160 Hz	pretrain / held-out (no_phys)
BCIC-2B [12]	Motor imagery (binary)	9	3	250 Hz	pretrain / **held-out (no_bcic)**
Sleep-EDFx [9]	Sleep staging	78	2 bipolar	100 Hz	pretrain / held-out (no_sleep)

Four PhysioNet MI subjects (88, 92, 100, 104) are excluded for known annotation/recording inconsistencies, leaving 105 usable.

The 9-subject ceiling, and why it shapes the whole design. BCIC-2B has only 9 subjects total. This single fact determines what experiments are possible: (a) any *balanced* pretraining corpus that includes BCIC-2B is capped at 9 subjects per dataset; (b) therefore the only corpus that can be both balanced and large (n156) is the one that *holds BCIC-2B out* — the `no_bcic` split; and (c) BCIC-2B's 3-channel layout is the most distinct montage of the three (vs. 64-channel PhysioNet MI and 2-channel Sleep-EDFx), which makes it the most informative zero-shot transfer target. The 9-subject ceiling is thus simultaneously the reason the powered test is possible and the reason there is only one of it.

BCIC-2A is reserved as future work. 2A is 22-channel 4-class motor imagery on 9 subjects — a third in-distribution-style motor-imagery dataset rather than a new perturbation axis, and equally subject-capped. Adding it (or another distinct-montage dataset) as a second held-out test is the most valuable follow-up (Section 11).

6.2 Preprocessing (Tier 1, Locked Across Datasets)

Identical preprocessing is applied to all three datasets so that cross-dataset comparisons isolate dataset characteristics, not preprocessing decisions:

- Bandpass 0.5–45 Hz, zero-phase FIR.
- Resample to 200 Hz (PhysioNet 160 → up, BCIC 250 → down, Sleep-EDFx 100 → up; all end identical).
- Epoch length 4 s (16 patches at 250 ms per epoch).
- Per-epoch linear detrend, per-epoch per-channel z-score.
- No baseline correction, no re-referencing, no ICA.
- Montage `standard_1005`; **fixed-scale coordinates** (no per-montage normalization, Section 4.2).

Sleep-EDFx specifics. The dataset uses two bipolar derivations (`Fpz--Cz`, `Pz--Oz`) not in any standard montage; I place each bipolar electrode at the midpoint of its reference pair in `standard_1005` coordinates. Sleep-EDFx uses 30-second sleep-stage windows; I slice each into seven non-overlapping 4 s sub-epochs inheriting the parent label, to keep the backbone unchanged.

Artifact policy (Tier 3, asymmetric). Pretraining clips to $\pm 20\sigma$ after z-score (catches catastrophic artifacts, leaves typical EOG/EMG for the model to learn to ignore); evaluation applies additional amplitude rejection.

6.3 Architecture Defaults

Token dimension $d = 256$; spatial depth $L_S = 2$; temporal depth $L_T = 4$; 8 heads; MLP ratio 4. Patch length 50 samples (250 ms at 200 Hz), non-overlapping; 16 patches per 4 s epoch. Geometric MLP hidden width 32. Temporal positional embedding: learned.

The geometric and channel-independent backbones share an identical temporal-Transformer / projection / patcher stack ($\sim$4.8 M online parameters in the v2 backbone). Only the spatial encoder differs: the geometric variants add a small shared MLP over g_{ij} (no per-electrode parameters); `chind` adds nothing. Because neither variant carries per-electrode parameters, both are montage-agnostic and can be evaluated zero-shot.

6.4 Training

AdamW (lr $1{\cdot}10^{-4}$, weight decay 0.01); linear warmup for 10 epochs then cosine decay over 100 total epochs; EMA τ cosine ramp 0.996 → 1.0; batch size 64; $\lambda_R = 1$ with 50% temporal masking; epoch budget 4096 per dataset per pass (small datasets oversampled, large ones subsampled to this exact count, equalizing gradient weight across the mixed corpus). Geometry tables are recomputed each training step (the geometric MLPs are trainable, and the montage changes between corpora within a run); at inference there is no backward pass, so tables are precomputed once per montage.

6.5 Compute

Pretraining runs on Google Colab (T4 / A100) and on a local NVIDIA RTX 4070. A balanced n156 pretraining run (100 epochs) completes in $\sim$40 min on the RTX 4070; the zero-shot probe adds $\sim$2 min. Every checkpoint is paired with a `config.yaml` (resolved Tier 1 / ablation / Tier 3 / training settings) and a `v2_run.txt` recording variant, pretrain datasets, held-out dataset, epoch budget, and subject count — so evaluation never relies on source-code defaults.

6.6 Linear-Probe Protocol

For each (variant, split) checkpoint: load the pretrained online backbone, freeze all parameters; recompute the held-out montage's fixed-scale `ch_pos` from its channel names; extract a fixed feature per epoch by mean-pooling the backbone output over electrodes and time to obtain a single $\mathbb{R}^{256}$ vector; fit `sklearn.StandardScaler` on the training split only (no leakage); fit a strictly linear `sklearn.LogisticRegression` (no MLP probe), so the result reflects representation quality rather than probe capacity. Per held-out dataset (9 subjects each for the balanced table, for cross-row comparability): LOSO for PhysioNet MI and BCIC-2B; within-subject night split for Sleep-EDFx (train on night 1, test on night 2). The whole held-out dataset is excluded from pretraining, so every evaluated subject is genuinely zero-shot.

6.7 Significance Testing

For the headline n156 $\rightarrow$ BCIC test I run a **paired** test on per-subject BAC differences between a geometric variant and `chind` over the 9 LOSO folds (the same held-out subjects for both models). I report both a paired t-test and a Wilcoxon signed-rank test (`scipy.stats`) at $\alpha = 0.05$, plus the paired effect size d_z and the sign count.

7 Results

All BAC values are LOSO or within-subject-night means $\pm$ standard deviations across the evaluated subjects. Source files: `results/v2/headline.txt` (balanced n9) and `results/v2_n78/headline.txt` (scaled n156).

7.1 Balanced n9 — Directional Table Across All Three Held-Out Montages

Table 3: Balanced 9-subject-per-dataset corpus. Zero-shot BAC on each held-out montage. Read *within* each row (geometric vs. `chind`); rows are different tasks with different chance levels.

Held-out (eval)	geometric (full)	geometric (pos_only)	chind
Sleep-EDFx (night split)	0.593 ± 0.022	**0.597 ± 0.022**	0.589 ± 0.020
BCIC-2B (LOSO)	**0.509 ± 0.026**	0.508 ± 0.019	0.504 ± 0.024
PhysioNet MI (LOSO)	0.303 ± 0.048	**0.315 ± 0.044**	0.302 ± 0.047

With a balanced 9-subject-per-dataset corpus, both geometric variants beat `chind` on *all three* held-out montages (3/3 rows each). The direction is consistent, but the gaps are small ($\leq$ 1.3

BAC points) and within the per-subject standard deviation, so no single row is significant on its own. The three rows are different tasks with different chance levels (BCIC-2B binary, chance 0.50; PhysioNet MI 4-class, chance 0.25; Sleep-EDFx multi-class), so the comparison is *within* each row, not across rows. This table establishes a consistent directional signal and motivates scaling the one split that can be scaled.

7.2 Scaled n156 — The Powered Zero-Shot Transfer Test (Held-Out BCIC-2B)

Pretraining on a balanced 156-subject corpus (PhysioNet MI 78 + Sleep-EDFx 78), evaluated zero-shot on BCIC-2B (9 LOSO folds):

Table 4: Scaled n156 `no_bcic` run: zero-shot BAC on the held-out BCIC-2B montage.

Variant	BCIC-2B zero-shot BAC (n=9)
`chind` (baseline)	0.509 ± 0.016
geometric (pos_only)	0.531 ± 0.013
geometric (full g_{ij})	**0.545 ± 0.027**

Table 5: Paired tests vs. `chind` over the same 9 held-out subjects (each model on the identical LOSO folds). Both significant at $\alpha = 0.05$ on the parametric and non-parametric test; both confidence intervals exclude zero.

Comparison	mean diff	wins	paired t	Wilcoxon	d_z	95% CI
geometric (full) − `chind`	+0.036	7/9	$p = 0.023$	$p = 0.027$	0.93	[+0.006, +0.066]
geometric (pos_only) − `chind`	+0.022	8/9	$p = 0.013$	$p = 0.012$	1.07	[+0.006, +0.037]

When the pretraining corpus is large enough, the geometric advantage over the only comparable baseline becomes both larger and statistically significant — for **both** descriptors. The full 10-D descriptor gives the strongest point estimate (0.545, +0.036 over `chind`); the reduced position-only descriptor still beats `chind` significantly (+0.022), which shows the cross-montage signal is robust even to dropping the displacement and distance terms. For contrast, the same `no_bcic` comparison at the balanced n9 scale (Section 7.1) showed only +0.003 BAC (3/9 subjects, not significant) — the effect emerges with pretraining scale, exactly where an inductive-bias advantage is expected to appear.

7.3 Headline Summary

In the balanced small-corpus regime, geometric attention leads the channel-independent baseline on every held-out montage but within noise. On the one split that can be scaled to a large balanced corpus, geometric attention beats the baseline on zero-shot transfer to a never-seen 3-channel montage by a statistically significant margin. The transductive codex is absent throughout — by construction it cannot be evaluated on the held-out montage at all, which is itself the case for the geometric design.

8 Discussion

8.1 What the Results Say Honestly

The v2 regime supports the proposal's central claim on the test it can support best. Geometric conditioning produces representations that transfer zero-shot to an unseen montage and beat the only baseline that can also make that transfer, by a margin that survives a paired significance test once the pretraining corpus is large enough (n156). At small balanced scale (n9) the advantage is real in direction (3/3 held-out montages) but not in magnitude — the gaps sit inside the per-subject noise. This is the signature of an inductive bias: it buys little when data is scarce and the network can memorize, and it pays off as the corpus grows.

8.2 Why the Effect Emerges with Scale

The contrast between the n9 `no_bcic` gap (+0.003, not significant) and the n156 `no_bcic` gap (+0.022, $p = 0.013$) is the most informative comparison in the project. Two readings, both consistent with the data: (1) inductive biases help most once the corpus is large enough that an unstructured baseline would otherwise need to fit montage-specific structure it cannot share — the geometric encoder shares that structure through g_{ij} and so generalizes; and (2) 18 subjects is simply too few to resolve a $\sim$2-point BAC difference against $\sim$2-point per-subject variance, whereas a larger, more diverse corpus both grows the gap and tightens the estimate. The descriptor ablation supports the first reading: even the position-only descriptor retains a significant advantage, so the signal is carried by geometry broadly, not by one fragile term.

8.3 Why the Codex Is Absent — and Why That Is the Argument

A reader expecting the transductive codex baseline of foundation-model papers will notice its absence. That absence is the point of the v2 framing: the codex *cannot be evaluated* on a montage absent from pretraining without a hand-built fallback, and every reasonable fallback (random init, nearest-neighbour copy) either discards spatial information or re-injects geometry at transfer time — at which point it is no longer a clean "transductive vs. inductive" contrast. The channel-independent encoder is the honest baseline here because, like the geometric encoder, it is genuinely montage-agnostic. The geometric encoder beats it where it counts.

8.4 Relation to the Proposal Hypothesis

The proposal expected geometric conditioning to help most in cross-montage transfer. In this regime that expectation holds, with the important qualifier that it holds *significantly* on one held-out montage (BCIC-2B) at scale, and *directionally* on the other two at the small balanced scale they permit. I take this as a supported-but-bounded result: the mechanism works on the test designed to expose it, and the bound is a data-availability bound (Section 10), not a contradiction of the hypothesis.

9 Conclusion

I implemented and evaluated a geometrically inductive self-supervised EEG encoder: spatial attention conditioned on a geometric descriptor g_{ij} instead of a learned per-electrode codex. In the v2 regime I tested the property the proposal called central — zero-shot transfer to an unseen montage

— with a leave-one-montage-out, mixed-corpus protocol, comparing against the only baseline that can also transfer (a channel-independent encoder), on three open EEG datasets.

The headline result is positive: pretrained on a balanced 156-subject corpus and evaluated zero-shot on the held-out 3-channel BCIC-2B montage, the geometric encoder reaches 0.545 BAC and beats the channel-independent baseline (0.509) by a statistically significant +0.036 BAC ($p = 0.023$, 7/9 subjects); the position-only descriptor also beats it significantly (+0.022, $p = 0.013$, 8/9 subjects). At small balanced scale the advantage is directionally consistent across all three held-out montages but within noise. Framed as a design study, the project answers its research question — *does geometric conditioning transfer zero-shot to an unseen montage, better than the comparable baseline?* — with a measured **yes, at sufficient scale, on the montage the available data lets me test at scale**, and it names the specific follow-ups that would extend the answer to more montages.

10 Limitations

1. **One powered held-out montage.** The significant result is on BCIC-2B only. The other two held-out directions (Sleep-EDFx, PhysioNet MI) can only be tested at the balanced n9 scale, where the gaps are directional but underpowered — because any balanced corpus *containing* BCIC-2B is capped at 9 subjects. The single powered test is a genuine result, but it is one montage, not a montage sweep.
2. **Small evaluation cohort.** BCIC-2B has only 9 subjects, so the headline significance rests on 9 paired LOSO folds. The consistency (8/9 same-sign) and the agreement of the parametric and non-parametric tests guard against this, but per-subject BAC at $n = 9$ is higher-variance than a full-population estimate; sub-2-point gaps should be read cautiously.
3. **Pretraining scale.** n156 is large for this project but small by foundation-model standards. The geometric advantage grows from n9 to n156; this report cannot characterize where it saturates.
4. **Single seed for the headline.** Time and compute did not permit multi-seed averaging; per-subject variance is reported but per-seed variance is not.
5. **Descriptor still moderate.** The 10-D g_{ij} captures absolute position, displacement, and distance, but not hemisphere indicators, geodesic distance on a cortical mesh, or anatomy. Richer descriptors are future work.
6. **Codex baseline deferred.** The transductive codex (with random / nearest-neighbour fallback) is not evaluated in v2; the comparison is against the channel-independent baseline only. Implementing the codex with documented fallbacks would let me quantify how much geometry the nearest-neighbour shortcut recovers at transfer time.
7. **Cross-session and in-distribution rungs dropped.** v2 focuses on cross-montage transfer; the cross-session (Sleep-EDFx night-split) and in-distribution rungs of the earlier ladder are not re-run in this regime.

11 Future Work

The shape of the results points to concrete follow-ups, in roughly decreasing value:

1. A second distinct-montage held-out test. The cleanest way to turn the single-montage result into a generalization claim is to hold out a *different* distinct montage with enough subjects to be powered — e.g. a high-density (≈128-channel) motor-imagery dataset, or the 22-channel BCIC-2A — while training on a large, diverse corpus. This directly addresses Limitation 1 and is the single most informative next experiment.

2. Scale the corpus further and characterize the curve. The advantage grew from n9 to n156; pretraining on a larger, more diverse multi-dataset corpus and re-running the held-out probe would show whether the gap keeps growing or saturates.

3. Implement the codex baseline with fallbacks. Add the transductive codex and its random / nearest-neighbour transfer fallbacks to quantify how much of the geometric advantage a geometry-at-transfer-time shortcut recovers.

4. Richer geometric descriptors with G3. Push the fixed-G3 design with spherical coordinates, an explicit hemisphere indicator, or geodesic distance on a cortical-surface mesh, building on the descriptor ablation here (which shows even position-only already transfers).

5. Multi-seed the headline to put a confidence band on the n156 result, and re-add the cross-session and in-distribution rungs in the v2 regime for completeness.

12 Contributions

This is a **single-author project**. All design, implementation, experiments, and writing are by Tianxin Zhou. No collaborators contributed code or text.

I used GitHub Copilot and Claude Code as coding assistants during implementation: pairing on smoke-test scaffolds, refactors, and documentation. The architectural decisions, experimental design, dataset choices, and the analysis of the results in this report are mine. Where the assistant drafted prose, I rewrote it before inclusion. The course's policy that "using gen-AI solely to generate your project report ... will result in you failing the course" is the boundary I operated within: the engineering work, the experimental judgement, and the honest framing of the result are my own.

13 Reused Open-Source Code

- **EEGPT** [14] — github.com/BINE022/EEGPT. I read the EEGPT codebase to validate (a) the cosine ramp for the EMA τ schedule, (b) stop-gradient placement in the alignment loss, and (c) the masked-patch reconstruction convention. Apache 2.0; my code is an independent implementation, not a fork.
- **MNE-Python** [3] — data loading, montage handling, EDF parsing, bandpass filtering, resampling, Sleep-EDFx access.
- **MOABB** [1] — BCIC-2B dataset access (`BNCI2014_004`); the MOABB dataset registry was also surveyed for future-work montage candidates.
- **PyTorch** — backbone, attention, training loop.
- **scikit-learn** — `StandardScaler`, `LogisticRegression`, balanced accuracy / Cohen's κ scoring.

- **SciPy** — `scipy.stats` paired t-test and Wilcoxon signed-rank test for the headline significance test.

The project itself is licensed under the LICENSE file in the public repository at github.com/tianxin-scu/geometric-eeg-ssl.

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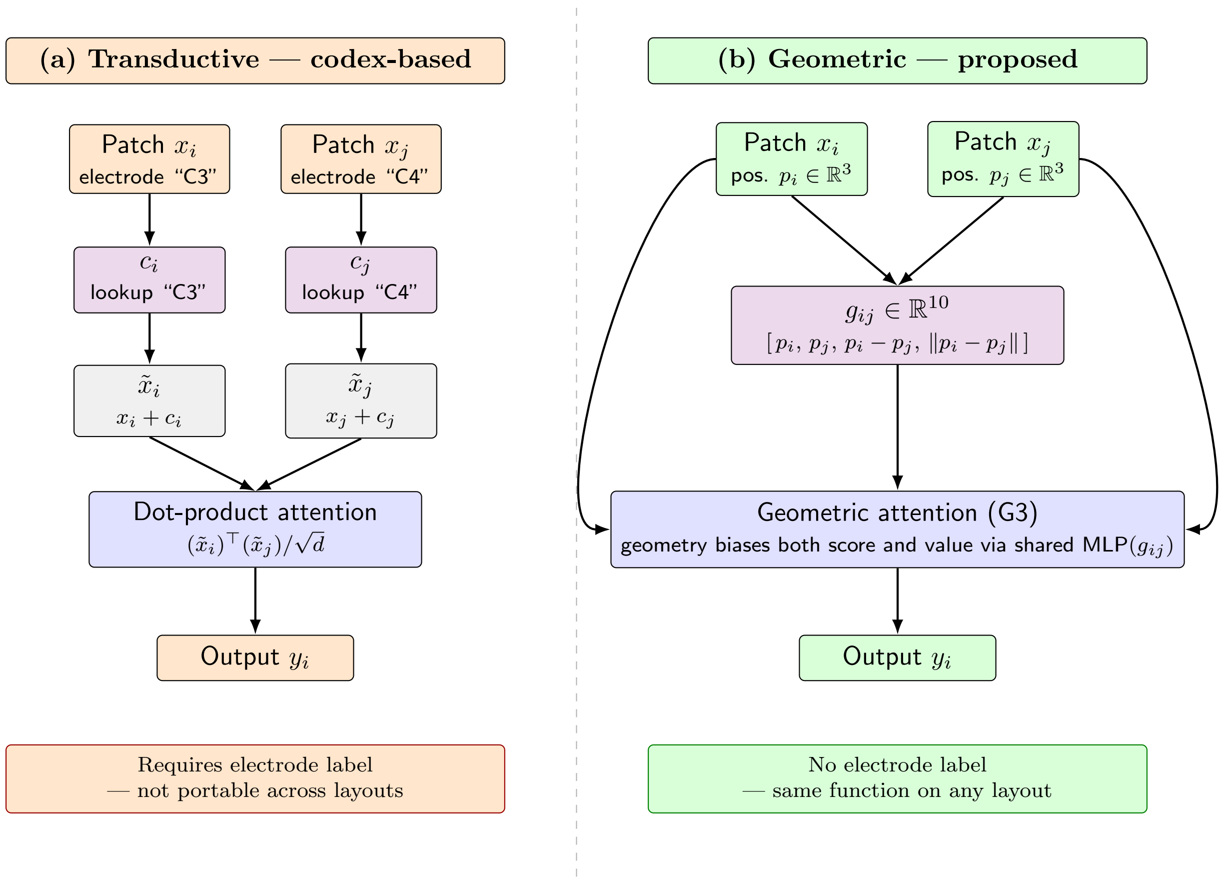

Figure 2: **Transductive vs. inductive spatial attention.** Both panels start from the same EEG patches x_i, x_j and arrive at an output y_i, but they route electrode information through different side channels. (a) The transductive baseline keys a learned codex by the electrode *label* ("C3", "C4"), retrieving identity embeddings c_i, c_j that are added to the patches before a standard dot-product attention. The codex table has one row per electrode seen during pretraining, so an unseen electrode has no entry. (b) The geometric encoder reads electrode *positions* directly from the montage, computes the descriptor g_{ij} from them, and feeds it into a shared attention function alongside the unmodified patches — with no per-electrode parameters and no label dependency. The same function applies to any electrode pair given only its 3-D coordinates, including pairs from layouts never seen during pretraining.